BCA Semester 4 Syllabus

ABOUT THIS DOCUMENT

SAMPLE / DUMMY DOCUMENT for testing only. Not real university policy.

BCA401 DATABASE MANAGEMENT SYSTEMS

Unit 1: Introduction to databases, data models, ER diagrams and the relational model.
Unit 2: SQL: DDL, DML, joins, subqueries and views.
Unit 3: Normalization: functional dependencies, 1NF, 2NF, 3NF and BCNF.
Unit 4: Transactions, ACID properties, concurrency control and recovery.
Unit 5: Introduction to NoSQL databases and indexing.
Textbook: Fundamentals of Database Systems (dummy reference). Credits: 4. Internal assessment 40 marks, end-semester exam 60 marks.

BCA402 PYTHON PROGRAMMING

Unit 1: Python basics, data types, operators and control flow.
Unit 2: Functions, modules, packages and exception handling.
Unit 3: Lists, tuples, dictionaries and sets; file handling.
Unit 4: Object oriented programming: classes, inheritance and polymorphism.
Unit 5: Libraries for data work: NumPy and pandas basics, and an introduction to web APIs.
Lab: 12 practical sessions in Computer Lab 1. Credits: 4.

BCA403 COMPUTER NETWORKS

Unit 1: Network models, OSI and TCP/IP layers.
Unit 2: Physical layer, data link layer and error detection.
Unit 3: Network layer, IP addressing, subnetting and routing algorithms.
Unit 4: Transport layer: TCP, UDP, congestion control.
Unit 5: Application layer protocols: HTTP, DNS, SMTP and network security basics. Credits: 3.